

Logica de ordinul I

Ex 18

$$M = (N, S, P) \quad , \quad S(x, y, z) \text{ "A"} \Leftrightarrow x + y = z$$

$$P(x, y, z) \text{ "A"} \Leftrightarrow xy = z$$

1) Se dă o serie de formule cu o variabilă x "A"

a) $x = 0$ $S(x, 0, 0)$
 $P(x, 1, 0)$

b) $x = 1$ $P(x, 1, 1)$
 $S(x, 0, 1)$

c) $x = 2$ $P(x, 1, 2)$
 $S(x, 0, 2)$

d) $x = \text{arbitrary}$ $\exists y P(2, y, x)$
 $\exists y S(y, y, x)$

e) $x = \text{nr impar}$

$$(2k+1) * (2k+1) = 4k^2 + 2 = 2(2k^2 + 1)$$

~~$\exists y S$~~

$$\neg (\exists y P(2, y, x))$$

$$\boxed{x = 2k+1} \Leftrightarrow \exists k \exists p (P(2, k, p) \wedge S(p, 1, x))$$

f) x este nr prim $\Rightarrow$ nu are divizorii

$$\neg (\exists y \exists z (P(y, z, x) \wedge P(y, 1, y) \wedge P(z, 1, z) \wedge S(1, x, x)))$$

$$\exists y S(y, 0, 1) \wedge P(x, 1, y)$$

$$(\forall y \forall z (P(y, z, x) \rightarrow (S(y, 0, 1) \vee S(y, 0, x)))) \wedge \neg S(x, 0, 0) \wedge \neg S(x, 0, 1)$$

2) Se da o serie de formule cu 2 variabile x, y „A”

a) $x = y$ $S(x, 0, y)$
 $P(x, 1, y)$

b) $x \leq y \Rightarrow \exists z S(x, z, y)$

c) $x < y \exists z S(x, z, y) \wedge \neg S(x, 0, y)$

d) $y \% x = 0$
 $\exists k P(x, k, y)$

e)

3) x, y, z libre

a) $z = \text{commune}(x, y)$

$$\exists k P(x, k, z) \wedge \underbrace{\exists q P(y, q, z)}_{\exists q} \wedge \cancel{S(k, z)} \wedge \cancel{S(q, z)}$$

$$\begin{cases} \int z dx = 0 \\ \int z dy = 0 \end{cases}$$

MIC

$\Rightarrow$ ~~the~~

~~the~~ ~~the~~ $z = \text{minimum}$

$$\wedge \forall d \left(\left(\exists k P(k, d, x) \wedge \exists q P(q, d, y) \right) \rightarrow \exists t P(t, d, z) \right)$$

b) $z = \text{commune}(x, y) \Leftrightarrow \int x/z \propto \int y/z$
 $\int \forall d (d/x \propto d/y) \rightarrow dz$

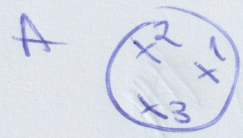
$z = \text{commune}(x, y) \Leftrightarrow \int x/z \propto \int y/z$
 $\int \forall m (x/m \propto y/m) \rightarrow z/m$

Cap 3. Teoria multimiilor

"naiv" O multime este o colecție de puncte obiecte bine determinate (unice) și care au o proprietate comună numită elem multimei.

O multime poate fi dată prin enumerarea elementelor $A = \{1, 2, 3\}$

2) diagrame Venn-Euler



3) printr-o prop comună

$$A = \{x \mid x \text{ este nr nat și } 1 \leq x \leq 3\}$$

$$B = \{M \mid M \notin M\} \rightarrow \text{paradoxul Russell}$$

$$\text{ex: } A = \{1, \underbrace{\{1, \{1, \dots\}}_A\}$$

$A \quad A \in A$

$B \in B?$ Dacă $NA \Rightarrow B \in B \Rightarrow B \notin B$ fals

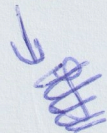
Dacă $NU \Rightarrow B \notin B \Rightarrow B \in B$ fals

Notatii: $\{ \}$ not $\emptyset$ multime vide

$$A \subseteq B \stackrel{\text{def}}{=} \forall x (x \in A \rightarrow x \in B)$$

$$A = B \stackrel{\text{def}}{=} \forall x (x \in A \leftrightarrow x \in B)$$

$$\mathcal{P}(M) = \{ S \mid S \subseteq M \} \text{ Dacă } |M| = n \text{ at } |\mathcal{P}(M)| = 2^n$$



(20) Fie $A, B, C \subseteq U$

$$*) A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Fie x arbitrar
(*)

$$x \in A \cap (B \cup C) \stackrel{\text{def "}\cap\text{"}}{=} x \in A \wedge x \in (B \cup C) \stackrel{\text{def "}\cup\text{"}}{=}$$

$$\Leftrightarrow x \in A \wedge (x \in B \vee x \in C) \Leftrightarrow$$

$$\stackrel{\text{distrib}}{=} (x \in A \wedge x \in B) \vee (x \in A \wedge x \in C)$$

pot să dev

$$\stackrel{\text{def "}\cap\text{"}}{=} (x \in (A \cap B)) \vee (x \in (A \cap C))$$

$$\stackrel{\text{def "}\cup\text{"}}{=} x \in (A \cap B) \cup (A \cap C) = A \cap (B \cup C)$$

(21) b) $A \setminus B = A \cap C_B$



Fie x arbitrar

$$x \in (A \setminus B) \stackrel{\text{def "}\setminus\text{"}}{=} x \in A \wedge x \notin B \stackrel{\text{def}}{=} x \in A \wedge \neg(x \in B)$$

$$\underline{x \in A \cap B} \quad \left| \begin{array}{l} \stackrel{\text{def}}{=} \\ \text{"}\cap\text{"} \end{array} \right. x \in A \wedge x \in B \stackrel{\text{def}}{=} "C"$$

$$\Leftrightarrow x \in A \wedge x \notin B$$

$$\Leftrightarrow x \in A \wedge \neg(x \in B)$$

g) De Morgan

$$1) \overline{C(A \cup B)} = \overline{C(A) \cap C(B)}$$

$$2) \overline{C(A \cap B)} = \overline{C(A) \cup C(B)}$$

1) Sei x beliebig, $x \in \overline{C(A \cup B)}$

$$\stackrel{\text{def "C"}}{=} x \notin (A \cup B)$$

$$\Leftrightarrow \neg x \in (A \cup B)$$

$$\stackrel{\text{not}}{=} \overline{x \in A \cup B}$$

$$\stackrel{\text{def "U"}}{=} \overline{x \in A \wedge x \in B}$$

De Morgan

$$\Leftrightarrow x \notin A \wedge x \notin B$$

$$\Leftrightarrow \overline{C(A) \cap C(B)}$$

Sei $x \in \overline{C(A) \cap C(B)}$

$$\stackrel{\text{def "C"}}{=} \overline{x \in A \wedge x \in B}$$